STAT 530 Bioinformatics

TTh 2:00 – 3:30 pm in 136 Loomis

Spring 2025

Course description

This course will teach general principles and skills for the analysis of genomic data. It will provide an framework for thinking about bioinformatics, introduce necessary concepts of biology, statistics, and AI-assisted programming. Students will then use these tools to replicate analyses on bulk, single-cell, and spatial transcriptomics data from a recently published paper. Students will also perform new analyses that go beyond those reported in the original publication. A final group project will test students' ability to independently carry out biological research.

Resources

Instructor:

- Name: Dave Zhao
- Office hours: Friday 2 pm – 3 pm
- Location: IGB 2406
- Email: sdzhao@illinois.edu

TA:

- Name: Bo Yuan
- Office hours: Mondays 11 am – 12 pm
- Location: IGB 2410
- Email: boy6@illinois.edu

Course materials: https://canvas.illinois.edu

- Submit assignments on Canvas page.
- Canvas page contains everything you need for this class.

Discord: https://discord.gg/ne8A5hx8

- Discuss with your classmates on Discord.
- You can collaborate on homeworks but please write up and submit your assignments independently.

Assessments

Homeworks

- Due Tuesdays at 2:00 pm; see schedule
- Two lowest scores will be dropped
- No late assignments will be accepted, including those that were late due to entering the course late, unless you have a DRES accommodation or a Verified Absence

Final project

- See schedule for due date
- You will be asked to perform a few analyses on the datasets from class that go beyond what is discussed in the lectures.

Grading

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- Homework: 85%, overall homework percentage will be the average of your percentages on each of the homeworks
- Final project: 15%

Converting final grade to letter grade:

A+$\in [97, 100]$	B+$\in [87, 90)$	C+$\in [77, 80)$	D+$\in [67, 70)$
A$\in [92, 97)$	B$\in [82, 87)$	C$\in [72, 77)$	D$\in [62, 67)$
A-$\in [90, 92)$	B-$\in [80, 82)$	C-$\in [70, 72)$	D-$\in [57, 62)$
F≤ 57			

Resources

These books may be helpful references.

1. "Molecular Biology of the Gene", Watson et al.
2. "Molecular Biology of the Cell", Albers et al.
3. "All of Statistics", Wasserman
4. "All of Nonparametric Statistics", Wasserman
5. "Structure and interpretation of computer programs", Abelson et al.
6. "The cartoon guide to genetics", Gonick and Wheelis

Note: some aspects of this syllabus may be subject to change. Any such changes will be announced and discussed in class.